# UNNS-U3-COR v001: Corrections and Strengthened Proofs for UNNS-U3 v001

UNNS Research Collective

10 November 2026

UNNS_U3_v001 is preserved unchanged as the original release. This supplement records three mathematical changes and one packaging fix.

## 1 U3.1b promoted to a theorem

**Theorem 1.** *For every $L \geq 1$ there is a lexicographically-first shortest path $w = (v_0, \ldots, v_L)$ with exactly $M(w) = 2^{L-1}$ compatible mark assignments (endpoints marked).*

*Proof.* $G$ is connected, infinite and locally finite, so the sphere of radius $L$ about any vertex is nonempty; take $w = P(a, b)$ with $d(a, b) = L$. A contiguous subpath of a lex-first shortest path is lex-first shortest between its endpoints: it is shortest, and a lexicographically smaller replacement would give a lexicographically smaller whole path of equal length. Hence every subset of the $L - 1$ internal vertices cuts $w$ into canonical routing segments, giving $2^{L-1}$ compatible markings; there cannot be more since only internal positions are free. □ (Computational check: subpath lemma and $M = 2^{L-1}$ confirmed for $L \leq 5$, three geodesics per length, source 1.) The "proof sketch / $L \leq 5$ only" qualification of v001 is removed.

## 2 Girth correction

v001 proved: $G$ is triangle-free and has odd girth 5. The assessment text suggested $\mathrm{girth}(G) = 5$; that is **false**. The cycle $2{-}8{-}36{-}9{-}2$ is a 4-cycle (edges $\{2, 8\}, \{8, 36\}, \{36, 9\}, \{9, 2\}$, each checked against the exact neighbour engine; it is the embedding already realized for $C_4$ in v001). Hence $\mathrm{girth}(G) = 4$, and odd girth $= 5$ by triangle-freeness plus the 5-cycle $6, 24, 96, 784, 27$. Self-contained triangle proof (checked symbolically): with $r_N = (N + 1)^2/N$ a triangle requires $r_a r_b = r_c$ (other sign patterns are the same equation after renaming), i.e. $c(a + 1)^2(b + 1)^2 = ab(c + 1)^2$. So $abc$ is a square, $h = \sqrt{abc}$, $c(a + 1)(b + 1) = (c + 1)h$, $c \mid h$, $h = ct$, $ab = ct^2$, $(a + 1)(b + 1) = t(c + 1)$, and $a + b = (t - 1)(1 - ct) \leq 0$, contradiction.

## 3 Entropy qualification

The v001 sentence "equality holds iff the roof is constant" is restricted. Nonconstant roofs cohomologous to a constant can also give equality; the general framework is pressure and equilibrium measures. What is established: in the golden-mean two-symbol family with roof values $(2, d, d)$, topological entropy equals the Parry-lift (Abramov) value exactly when $d = 2$ (6 of 56 embeddings; strict inequality in the other 50).

## 4 Packaging

v001's SHA256SUMS listed itself with the hash of an empty file. Manifests here exclude themselves. Original v001 substantive results are otherwise unchanged. Open obligations carry forward to U4: exact SELECT-I, FIELD-I and GIG definitions.